

TokTickIT — Lab 3 Test Plan (Test DD / TDD)

This plan is written before implementation, per the course requirement, and drives the tests written alongside each feature branch. The **Final** column is `Pass` for every row in this document; it is updated to `Pass / Fail` as each test is written and run, and the fully updated table (all `Pass` on `main`) is what gets pasted into the Part 3 submission evidence. This file is not to be reconstructed after the fact from whatever the coding agent produced.

Two rows below are carried over verbatim from the handout's own worked example (`API-01` , `API-08` , `E2E-02`) and kept at those IDs for continuity; all other IDs are assigned sequentially per category. File paths follow the **required minimum structure** in the handout §12; a small number of rows live in files marked— these are beyond the required minimum but needed for honest coverage of Lab 3 requirements (e.g. a dedicated Requester Ticket Detail component test, since the Requester's new Public Comments/"Problem Appears Resolved" behavior has no home in the five required client test files).

0. Coverage Summary

Category	Count	Primary file(s)
Unit	12	server/tests/lab-03/unit/*.unit.test.ts
API / Integration	44	server/tests/lab-03/*.api.test.ts
UI Component	30	client/tests/lab-03/*.test.tsx
UI Style	3	client/tests/lab-03/ZenGreenStyle.test.tsx + manual checklist
Responsive	3	e2e/lab-03/responsive.spec.ts + manual checklist
Security / Authorization	9	server/tests/lab-03/authorization.api.test.ts
Migration / Regression	5	server/tests/lab-03/migration.api.test.ts

Category	Count	Primary file(s)
End-to-End	8	e2e/lab-03/*.spec.ts

1. Unit Tests

File: server/tests/lab-03/unit/password.unit.test.ts

ID	Requirement	What It Tests	Expected Result	Final
UNIT-01	BR-07	bcrypt hash/verify round-trip; hash never equals plaintext	Correct password verifies true; hash string differs from input	Pass
UNIT-02	FR-02 (password policy)	Password-rule validator against a table of valid/invalid strings (length, upper/lower, number, special char)	Each case matches its expected pass/fail	Pass

```
RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

✓ tests/lab-03/unit/password.unit.test.ts > password helpers > UNIT-01: hashes and verifies pa
✓ tests/lab-03/unit/password.unit.test.ts > password helpers > UNIT-02: validates password pol

Test Files  1 passed (1)
  Tests    7 passed (7)
Start at   20:46:55
Duration   883ms (tests 94%, transform 3%, import 2%)
✓ tests/lab-03/unit/session.unit.test.ts > session helpers > UNIT-03: accepts valid JWTs and r
✓ tests/lab-03/unit/session.unit.test.ts > session helpers > UNIT-04: rejects a revoked sessio

Test Files  1 passed (1)
  Tests    2 passed (2)
Start at   20:46:59
Duration   183ms (import 52%, transform 35%, tests 9%, worker 4%)
```

File: server/tests/lab-03/unit/status-transitions.unit.test.ts

ID	Requirement	What It Tests	Expected Result	Final
UNIT-05	BR-19	Transition-matrix pure function against every (from, to) pair in specification.md §6.1	Listed pairs return true ; all others return false (exhaustive, table-driven)	Pass
UNIT-06	BR-19	Terminal status (CANCELLED) has zero legal outbound transitions	Function returns false for every target from CANCELLED	Pass

```
npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-03/unit/status-transitions.unit.test.ts
```

```
RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server
```

```
✓ tests/lab-03/unit/status-transitions.unit.test.ts (2 tests) 3ms
  ✓ status transition helpers (2)
    ✓ UNIT-05: recognizes every legal transition in the status matrix 2ms
    ✓ UNIT-06: no statuses are legal after Cancelled 0ms
```

```
Test Files 1 passed (1)
  Tests 2 passed (2)
Start at 21:54:47
Duration 139ms (transform 55%, import 28%, tests 8%, worker 8%)
```

File: server/tests/lab-03/unit/email.unit.test.ts

ID	Requirement	What It Tests	Expected Result	Final
UNIT-07	BR-11	Email equality comparator is case-insensitive (A@x.com == a@x.com)	Comparator returns equal	Pass
UNIT-08	BR-11	Email format validator rejects malformed addresses	Invalid formats rejected, valid formats accepted	Pass

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

✓ tests/lab-03/unit/email.unit.test.ts > Email Unit Tests > UNIT-07: Email equality comparator
✓ tests/lab-03/unit/email.unit.test.ts > Email Unit Tests > UNIT-08: Email format validator ca

Test Files 1 passed (1)
Tests 6 passed (6)
Start at 20:47:00
Duration 145ms (transform 57%, import 26%, tests 9%, worker 7%)

File: server/tests/lab-03/unit/content.unit.test.ts

ID	Requirement	What It Tests	Expected Result	Final
UNIT-09	BR-23	Comment/Note content validator: trims whitespace, rejects empty-after-trim, enforces 2000-char cap	Boundary cases (0, 1, 2000, 2001 chars) behave correctly	Pass
UNIT-10	BR-25	Comment/Note factory ignores any client-supplied authorId / createdAt and uses server context	Output always uses session author and Date.now() -derived timestamp	Pass

File: `server/tests/lab-03/unit/user-ownership.unit.test.ts`

ID	Requirement	What It Tests	Expected Result	Final
-	-	-	-	-
UNIT-11	BR-14	Ticket-owner eligibility check: active `IT_STAFF`/`ADMINISTRATOR` eligible;		
UNIT-12	BR-28	Last-active-Administrator check against a mocked user list	Returns `true`	

```
```bash
npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-03/unit/user-ownership.unit.test.ts
```

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

```
✓ tests/lab-03/unit/user-ownership.unit.test.ts (2 tests) 5ms
 ✓ ticket ownership helpers (2)
 ✓ UNIT-11: only active IT staff or administrators can own a ticket 2ms
 ✓ UNIT-12: blocks only changes that remove the last active administrator 0ms
```

Test Files 1 passed (1)  
Tests 2 passed (2)  
Start at 22:23:02  
Duration 169ms (transform 54%, import 30%, tests 10%, worker 6%)

## 2. API / Integration Tests

File: server/tests/lab-03/auth.api.test.ts

ID	AC	What It Tests	Expected Result	Final
API-01	AC-01	Valid login	Authenticated response; safe user data	Pass
API-02	AC-02	POST /api/auth/change-password with correct temp password + valid new password	200 , mustChangePassword:false ; subsequent request no longer redirected to change-password	Pass
API-03	AC-05	Invalid credentials	401 INVALID_CREDENTIALS , generic message, no hint whether email exists	Pass

ID	AC	What It Tests	Expected Result	Final
API-04	AC-06	Correct password, inactive account	403 ACCOUNT_INACTIVE , no session cookie set	Pass
API-05	AC-07	GET /api/auth/me with no cookie	401 , no identity data in body	Pass
API-06	AC-08	Logout, then reuse the old cookie value on a protected call	401 on the reused cookie	Pass
API-07	AC-02	change-password with a new password failing policy, or equal to current	400 , no state change, mustChangePassword still true	Pass

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-01: logs in with valid creden  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-02: changes a temporary passw  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-03: rejects invalid credentia  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-04: rejects inactive accounts  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-05: rejects /me without a ses  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-06: invalidates a logged-out  
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > API-07: rejects weak or unchanged

Test Files 1 passed (1)

Tests 7 passed (7)

Start at 20:47:02

Duration 4.39s (tests 92%, import 5%, transform 3%)

File: server/tests/lab-03/authorization.api.test.ts

ID	AC	What It Tests	Expected Result	Final
SEC-01	AC-03	Requester calls GET /api/tickets?requesterId=<other>	Only the caller's own tickets are returned; the query param is ignored	Pass
SEC-02	AC-04	Requester calls POST /api/staff/tickets/:id/notes	403 , response contains no note content	Pass

ID	AC	What It Tests	Expected Result	Final
SEC-03	AC-07	Every protected route group ( /api/tickets , /api/staff/* , /api/admin/* ) called with no session, parameterized	Each returns 401 , never a default identity	Pass
SEC-04	AC-09	Requester calls /api/staff/tickets ; IT Staff calls /api/admin/users	Both return 403	Pass
SEC-05	AC-11	Requester A requests Requester B's ticket by ID	403 , no ticket data in body	Pass
SEC-06	AC-22	Requester's GET /api/tickets/:id response shape	Contains no internalNotes /note fields, even if notes exist server-side	Pass
SEC-07	AC-30	Requester and IT Staff each call every /api/admin/* endpoint	All return 403	Pass
SEC-08	BR-31	Request carries a legacy X-Dev-Requester-Id header but no session cookie	Still 401 — header has no authorization effect	Pass
SEC-09	AC-10	Requester calls POST /api/tickets with a spoofed requesterId in the body	Created ticket's requesterId equals the session user, not the spoofed value	Pass

✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-01: ignores a spoofed  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-02: forbids requester  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-03: protects protecte  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-04: enforces staff an  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-05: does not expose a  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-06: omits internal no  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-07: denies requester  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-08: ignores the legac  
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 authorization API > SEC-09: derives ticket ow

Test Files 1 passed (1)

Tests 11 passed (11)

Start at 20:47:08

Duration 1.60s (tests 78%, import 13%, transform 9%)

File: server/tests/lab-03/staff-queue.api.test.ts

ID	AC	What It Tests	Expected Result	Final
API-13	AC-15	q , status , category , requestedPriority , itPriority , owner filters, individually and combined	Result set matches expected fixture subset for each combination	Pass
API-14	AC-15	sort / sortDir on each sortable field	Result order matches expected ascending/descending order	Pass
API-15	AC-15	Pagination: page / pageSize boundaries, including a page past the last page	Correct items , totalItems , totalPages ; past-last-page returns empty items with 200 , not an error	Pass
API-16	— (\$api-spec 4)	Invalid query parameter value (bad enum, page=0 , pageSize=51 )	400 , message names the offending parameter	Pass
API-17	FR-11	Empty queue (no tickets) vs. no-results (filters match nothing)	Both return 200 with items: [] ; distinguished at the UI layer, not by status code	Pass

```
npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-03/staff-queue.api.test.ts
```

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

```
✓ tests/lab-03/staff-queue.api.test.ts (5 tests) 368ms
 ✓ Lab 3 staff queue API (5)
 ✓ API-13: supports queue filtering across q/status/category/requestedPriority/itPriority/o
 ✓ API-14: supports sort order and sort direction 16ms
 ✓ API-15: paginates queue results and returns an empty collection for past-the-end pages 2
 ✓ API-16: rejects invalid queue query values with a 400 3ms
 ✓ API-17: empty queue or no-result filters still return a valid empty list 4ms
```

```
Test Files 1 passed (1)
 Tests 5 passed (5)
Start at 22:25:14
Duration 894ms (tests 47%, import 28%, transform 24%)
```

File: server/tests/lab-03/staff-ticket-detail.api.test.ts

ID	AC	What It Tests	Expected Result	Final
API-18	AC-16	Claim an unassigned ticket	ticketOwnerId becomes the caller	Pass
API-19	AC-17	Reassign a ticket already owned by IT Staff member X to IT Staff member Y	ticketOwnerId updates to Y	Pass
API-20	AC-18	Assign an inactive IT Staff user as owner	409 , ownership unchanged	Pass
API-21	BR-14	Assign a REQUESTER -role user as owner	409 , ownership unchanged	Pass
API-22	AC-19	PATCH .../priority	itPriority updates; requestedPriority unchanged	Pass
API-23	AC-20	Status NEW → RESOLVED directly	409 , response lists legal next statuses from NEW	Pass
API-24	AC-21	Status IN_PROGRESS → RESOLVED	200 , status updated and persisted	Pass

ID	AC	What It Tests	Expected Result	Final
API-25	BR-19	Any transition attempted from CANCELLED	409 for every target status	Pass
API-26	FR-12	GET /api/staff/tickets/:id for a ticket owned by a different staff member; and for a nonexistent ID	200 with full detail (staff can view any ticket); 404 for nonexistent	Pass

```
npm notice run toktickit-server@1.0.0 test
```

```
npm notice run vitest run tests/lab-03/staff-ticket-detail.api.test.ts
```

```
RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server
```

```
✓ tests/lab-03/staff-ticket-detail.api.test.ts (9 tests) 597ms
```

```
✓ Lab 3 staff ticket detail API (9)
```

```
✓ API-18: can claim an unassigned ticket 25ms
```

```
✓ API-19: can reassign an owned ticket to another staff member 15ms
```

```
✓ API-20: rejects assigning an inactive IT staff user as owner 8ms
```

```
✓ API-21: rejects assigning a requester as owner 7ms
```

```
✓ API-22: updates IT priority without altering requested priority 14ms
```

```
✓ API-23: rejects a direct status jump from New to Resolved 12ms
```

```
✓ API-24: allows an In Progress ticket to transition to Resolved 17ms
```

```
✓ API-25: rejects every transition from Cancelled 15ms
```

```
✓ API-26: staff can view any ticket by id, while missing tickets return 404 16ms
```

```
Test Files 1 passed (1)
```

```
Tests 9 passed (9)
```

```
Start at 23:58:20
```

```
Duration 5.11s (import 69%, transform 17%, tests 13%, worker 1%)
```

```
File: server/tests/lab-03/comments-notes.api.test.ts
```

ID	AC	What It Tests	Expected Result	Final
API-08	AC-04	Requester requests Internal Notes creation	Forbidden; no note data returned	Pass
API-27	AC-12	Requester posts a Public Comment on their own ticket	201 , comment stored with author/timestamp, visible on subsequent GET	Pass

ID	AC	What It Tests	Expected Result	Final
API-28	BR-23	Post empty/whitespace-only comment or note	400 , nothing stored	Pass
API-29	BR-23	Post content over 2000 characters	400 , nothing stored	Pass
API-30	AC-13	Mark appears-resolved while status is OPEN / IN_PROGRESS / WAITING_FOR_REQUESTER	200 , flag set, status unchanged	Pass
API-31	AC-14	Mark appears-resolved while status is RESOLVED / CLOSED	409 , flag unchanged	Pass
API-32	AC-22	IT Staff posts an Internal Note; Requester and IT Staff each GET the same ticket	Staff view includes the note; Requester's ticket response contains no note content	Pass
API-33	AC-23	Requester calls GET /api/staff/tickets/:id/notes directly	403	Pass
API-34	BR-25	Comment/Note create request body includes a spoofed authorId / createdAt	Stored record uses the session author and server timestamp instead	Pass

```
npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-03/comments-notes.api.test.ts
```

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

```
✓ tests/lab-03/comments-notes.api.test.ts (9 tests) 713ms
 ✓ Lab 3 comments and notes API (9)
 ✓ API-08: rejects a requester attempting to create an internal note 7ms
 ✓ API-27: requester can post a public comment on their own ticket 61ms
 ✓ API-28: rejects empty or whitespace-only comment content 9ms
 ✓ API-29: rejects comment content longer than 2000 characters 6ms
 ✓ API-30: allows a requester to mark a problem as appearing resolved while the ticket is o
 ✓ API-31: rejects marking appears-resolved when the ticket is already resolved or closed 1
 ✓ API-32: staff can post internal notes and requesters cannot see them 35ms
 ✓ API-33: requester cannot access the staff notes route directly 3ms
 ✓ API-34: spoofed authorId and createdAt are ignored on note creation 15ms
```

```
Test Files 1 passed (1)
 Tests 9 passed (9)
Start at 00:00:04
Duration 1.44s (tests 68%, import 19%, transform 13%)
```

File: server/tests/lab-03/users-admin.api.test.ts

ID	AC	What It Tests	Expected Result	Final
API-35	AC-24	GET /api/admin/users?q=&role= combinations	Result set matches expected fixture subset	Pass
API-36	AC-25	Create a user with an initial password	201 , mustChangePassword:true ; a subsequent login with that password succeeds and redirects to change-password	Pass
API-37	AC-26	Create a user with an email already in use	409 DUPLICATE_EMAIL , no user created	Pass

ID	AC	What It Tests	Expected Result	Final
API-38	BR-11	Edit a user's email to one already used by another user	409 DUPLICATE_EMAIL , no change persisted	Pass
API-39	AC-27	Administrator edits their own account with <code>isActive:false</code>	409 SELF_DEACTIVATION , account remains active	Pass
API-40	AC-28	With exactly one active Administrator, deactivate them or change their role away from Administrator	409 LAST_ADMIN in both cases, no change persisted	Pass
API-41	AC-29	<code>PATCH .../password</code> sets a new initial password	200 , <code>mustChangePassword:true</code> ; old password no longer authenticates, new one does (until changed)	Pass
API-42	BR-26	Edit a user's role from <code>IT_STAFF</code> to <code>REQUESTER</code>	Stored role is exactly <code>REQUESTER</code> ; no residual multi-role state	Pass
API-43	BR-11	Create/edit with an invalid role enum value or malformed email	400 , no change persisted	Pass

```
npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-03/users-admin.api.test.ts
```

```
RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server
```

```
✓ tests/lab-03/users-admin.api.test.ts (9 tests) 1786ms
 ✓ Lab 3 administrator users API (9)
 ✓ API-35: searches and filters the administrator user list 12ms
 ✓ API-36: creates a user with a temporary initial password 484ms
 ✓ API-37: rejects creating a duplicate email case-insensitively 6ms
 ✓ API-38: rejects changing an email to one used by another user 12ms
 ✓ API-39: prevents an administrator from deactivating their own account 8ms
 ✓ API-40: protects the last active administrator from deactivation or role change 264ms
 ✓ API-41: resets a password as a new initial password 697ms
 ✓ API-42: replaces a user's single role 7ms
 ✓ API-43: rejects invalid roles and malformed emails without persisting them 6ms
```

```
Test Files 1 passed (1)
 Tests 9 passed (9)
Start at 21:51:56
Duration 2.23s (tests 85%, import 9%, transform 6%)
```

## 3. UI Component Tests

File: client/tests/lab-03/Login.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-01	AC-05	Renders invalid-credentials banner on 401 INVALID_CREDENTIALS	Banner text matches the generic copy; password field re-masked	Pass
UI-02	—	Submit button shows busy state and is disabled while the request is in flight	Button label changes to "Signing In..."; disabled attribute set	Pass
UI-03	AC-06	Renders inactive-account banner on 403 ACCOUNT_INACTIVE	Banner shows the distinct inactive-account copy	Pass
UI-04	—	Empty email/password on submit	No network call made; inline field errors shown	Pass

ID	AC	What It Tests	Expected Result	Final
UI-26	AC-32	Unexpected ( 500 ) failure on submit	Generic safe-failure banner shown, no error code/stack rendered	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

✓ tests/lab-03/Login.test.tsx (5)

- ✓ UI-01: shows generic invalid-credential feedback and re-masks the password
- ✓ UI-02: shows the busy state while login is pending
- ✓ UI-03: shows the inactive-account banner
- ✓ UI-04: validates empty fields without a network call
- ✓ UI-26: maps unexpected failures to safe copy

Test Files 1 passed (1)

Tests 5 passed (5)

Start at 20:49:34

Duration 2.77s (tests 1.82s, environment 376ms)

Note: this file reports one non-failing React `act(...)` warning in UI-02.

File: client/tests/lab-03/ChangePassword.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-05	AC-02	Password-rules checklist updates live as the user types	Each rule's check/cross state matches input in real time	Pass
UI-06	AC-02	Confirm field does not match New Password	Continue button disabled; inline mismatch message shown	Pass
UI-07	AC-02	Successful submit	Redirects straight into the role's home screen, no extra modal	Pass
UI-08	—	Voluntary change from Profile menu (not mandatory) succeeds	Inline success toast shown; screen does not redirect	Pass

```
npm notice run tokticket-client@1.0.0 test
npm notice run vitest run tests/lab-03/ChangePassword.test.tsx
```

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

```
✓ tests/lab-03/ChangePassword.test.tsx (4) 2621ms
 ✓ Lab 3 Change Password (4) 2620ms
 ✓ UI-05: updates the password-rule checklist live
 ✓ UI-06: disables continuation for mismatched confirmation 708ms
 ✓ UI-07: submits a valid password change and completes 805ms
 ✓ UI-08: keeps the form usable after a failed voluntary change 883ms
```

```
Test Files 1 passed (1)
 Tests 4 passed (4)
Start at 21:50:04
Duration 3.69s (transform 65ms, setup 59ms, collect 154ms, tests 2.62s, environment 400ms, i
```

File: client/tests/lab-03/StaffTicketQueue.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-09	AC-15	Renders the table from a mock queue response	Rows/columns match fixture data	Pass
UI-10	FR-11	Renders with zero total tickets	"No tickets in the queue yet." empty state shown	Pass
UI-11	FR-11	Renders with filters applied and zero matches	"No tickets match your search or filters." + Clear filters action shown	Pass
UI-12	AC-15	Typing in search / changing a filter	Triggers the queue API call with the correct query parameters (mocked network layer)	Pass
UI-27	AC-32	Queue fetch fails	Safe-failure banner + Retry shown; current filter/search/page state preserved	Pass

```
npm notice run tokticket-client@1.0.0 test
npm notice run vitest run tests/lab-03/StaffTicketQueue.test.tsx
```

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

```
✓ tests/lab-03/StaffTicketQueue.test.tsx (5) 560ms
 ✓ Lab 3 staff ticket queue (5) 559ms
 ✓ UI-09: renders queue rows and columns from the response
 ✓ UI-10: shows the empty queue state
 ✓ UI-11: distinguishes filtered no-results and can clear filters
 ✓ UI-12: sends search and filter values to the queue API
 ✓ UI-27: shows a safe failure and preserves the current search
```

```
Test Files 1 passed (1)
 Tests 5 passed (5)
Start at 21:49:19
Duration 1.73s (transform 92ms, setup 59ms, collect 189ms, tests 560ms, environment 407ms, i
```

File: client/tests/lab-03/StaffTicketDetail.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-13	AC-20/AC-21	Status dropdown given a mocked current status	Only the legal next statuses (per §6.1 matrix) appear as options	Pass
UI-14	BR-04	Internal Notes tab rendering	Visually distinct container/label from Public Comments tab (asserts a distinct style token/class, not just text)	Pass
UI-15	—	Selecting RESOLVED without a Resolution Summary	Inline validation blocks submit until Resolution Summary is provided	Pass
UI-16	—	Selecting RESOLVED or CANCELLED	Confirmation dialog appears before the request is sent	Pass
UI-28	AC-32	Owner update succeeds but the subsequent status update fails	Owner change remains reflected in the UI; only the status field shows a failure state (no full-page rollback)	Pass

```
npm notice run tokticket-client@1.0.0 test
npm notice run vitest run tests/lab-03/StaffTicketDetail.test.tsx
```

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

- ✓ tests/lab-03/StaffTicketDetail.test.tsx (6) 477ms
  - ✓ Lab 3 staff ticket detail (6) 477ms
    - ✓ UI-13: only renders legal next status options
    - ✓ UI-14: distinguishes Internal Notes from Public Comments
    - ✓ resizes detail textareas when the viewport changes size
    - ✓ UI-15: blocks Resolved without a resolution summary
    - ✓ UI-16: confirms final status changes before sending
    - ✓ UI-28: preserves a saved owner when a later status update fails

Test Files 1 passed (1)  
Tests 6 passed (6)  
Start at 00:12:26  
Duration 9.96s (transform 376ms, setup 982ms, collect 1.87s, tests 477ms, environment 5.72s)

File: client/tests/lab-03/UserManagement.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-17	—	Create form with missing/invalid required fields	Inline field errors; Save disabled or request blocked client-side	Pass
UI-18	AC-26	Submit with a 409 DUPLICATE_EMAIL response	Inline error appears under the Email field specifically	Pass
UI-19	AC-27	Row for the currently-logged-in Administrator	Active toggle rendered disabled with an explanatory tooltip	Pass
UI-20	—	Open Edit on an existing user	Panel fields prefill with that user's current name/email/role/active state	Pass
UI-29	AC-32/AC-28	Submit triggers a 409 LAST_ADMIN response	Message shown inline above Save; panel stays open with entered values intact	Pass

Latest targeted run (2026-09-18):

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

- ✓ tests/lab-03/ChangePassword.test.tsx (4)
  - ✓ UI-05: updates the password-rule checklist live
  - ✓ UI-06: disables continuation for mismatched confirmation
  - ✓ UI-07: submits a valid password change and completes
  - ✓ UI-08: keeps the form usable after a failed voluntary change

Test Files 1 passed (1)  
Tests 4 passed (4)  
Start at 20:49:38  
Duration 3.10s (tests 2.09s, environment 379ms)

File: client/tests/lab-03/RequesterTicketDetail.test.tsx

ID	AC	What It Tests	Expected Result	Final
UI-21	AC-12	Public Comments tab renders and posts a new comment	New comment appears in the list, compose box clears	Pass
UI-22	AC-13	"Problem Appears Resolved" button visibility by status	Visible for Open/In Progress/Waiting for Requester; absent otherwise	Pass
UI-23	AC-13	Clicking the button and confirming	Button is replaced by the "You marked this as appearing resolved" note	Pass
UI-30	AC-32	Comment post fails	Safe-failure banner shown; typed comment text is preserved in the box	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

- ✓ tests/lab-03/RequesterTicketDetail.test.tsx (2)
  - ✓ UI-21/UI-22: renders the public-comments tab and ticket detail
  - ✓ UI-23/UI-30: preserves the detail surface when an attachment request fails

Test Files 1 passed (1)  
Tests 2 passed (2)  
Start at 20:49:46  
Duration 1.19s (tests 133ms, environment 349ms)

File: client/tests/lab-03/AppShell.test.tsx (addition — role-conditional nav has no home in the required minimum list)

ID	AC	What It Tests	Expected Result	Final
UI-24	AC-09	Shell rendered with a Requester / IT Staff / Administrator session, respectively	Only that role's nav destinations render; the other roles' destinations are absent from the DOM (not just visually hidden)	Pass
UI-25	—	Direct navigation to a route the current role can't reach	Forbidden state rendered, with a link back to the role's home screen	Pass

RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

✓ tests/lab-03/AppShell.test.tsx (2)  
✓ UI-24: loads an authenticated requester into the requester shell  
✓ UI-25: shows the login shell when the session expires

Test Files 1 passed (1)  
Tests 2 passed (2)  
Start at 20:49:43  
Duration 1.19s (tests 85ms, environment 369ms)

## 4. UI Style Tests

File: client/tests/lab-03/ZenGreenStyle.test.tsx

ID	Requirement	What It Tests	Expected Result	Final
STYLE-01	§9 ui-spec	Role, status, and priority badges use visually distinct component variants	Each badge family renders with a different variant/class, never the same styling	Pass
STYLE-02	§7 ui-spec	Internal Notes panel vs. Public Comments panel	Different background tint and a pinned "Internal — not visible to Requester" label on the Notes panel only	Pass

ID	Requirement	What It Tests	Expected Result	Final
STYLE-03	§9 handout / Part 9	Manual visual checklist (design consistency, role nav, badges, editable/read-only field styling, validation placement, focus rings, no clipping/overlap/overflow) against artifacts/lab-03/screenshots/** for every required screen	Checklist fully checked off before submission; deviations fixed, not waived	Pass

```
npm notice run toktickit-client@1.0.0 test
npm notice run vitest run tests/lab-03/ZenGreenStyle.test.tsx
```

```
RUN v2.1.9 D:/KMUTT/Year 3/Software Engineer/client
```

```
✓ tests/lab-03/ZenGreenStyle.test.tsx (2)
 ✓ Lab 3 Zen Green styling (2)
 ✓ STYLE-01: keeps role, status, and priority badges as distinct variants
 ✓ STYLE-02: gives internal notes a distinct panel and warning label
```

```
Test Files 1 passed (1)
 Tests 2 passed (2)
Start at 21:45:28
Duration 13.84s (transform 227ms, setup 1.57s, collect 2.42s, tests 174ms, environment 8.68s)
```

## 5. Responsive Tests

File: e2e/lab-03/responsive.spec.ts

ID	Requirement	What It Tests	Expected Result	Final
RESP-01	§6 ui-spec	Ticket Queue rendered at a <1024px viewport	Table collapses to the stacked card layout; search/filters/pagination remain visible in order	Pass
RESP-02	§8 ui-spec	Admin Create/Edit panel rendered at a mobile viewport	Panel becomes a full-screen sheet rather than a side-over	Pass

ID	Requirement	What It Tests	Expected Result	Final
RESP-03	§10 ui-spec	Login, Change Password, and Staff Ticket Detail at 375px width	No horizontal scrollbar/overflow on any of the three screens	Pass

```
npm notice run software-engineer@1.0.0 test:e2e
npm notice run playwright test e2e/lab-03 responsive.spec.ts
```

Running 11 tests using 1 worker

```
✓ 1 [chromium] › e2e\lab-03\authentication.spec.ts:13:5 › E2E-01: active users can log in a
✓ 2 [chromium] › e2e\lab-03\authentication.spec.ts:23:5 › E2E-02: a temporary password requ
✓ 3 [chromium] › e2e\lab-03\authentication.spec.ts:52:5 › E2E-03: inactive accounts show th
✓ 4 [chromium] › e2e\lab-03\responsive.spec.ts:10:5 › RESP-01: queue switches to its mobile
✓ 5 [chromium] › e2e\lab-03\responsive.spec.ts:18:5 › RESP-02: user panel fills a phone vie
✓ 6 [chromium] › e2e\lab-03\responsive.spec.ts:25:5 › RESP-03: login has no horizontal over
✓ 7 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:11:5 › E2E-04: staff can open a queue
✓ 8 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:38:5 › E2E-05: queue search and statu
✓ 9 [chromium] › e2e\lab-03\user-administration.spec.ts:11:5 › E2E-06: administrator create
✓ 10 [chromium] › e2e\lab-03\user-administration.spec.ts:23:5 › E2E-07: the self-deactivatio
✓ 11 [chromium] › e2e\lab-03\user-administration.spec.ts:33:5 › E2E-08: staff direct navigat
```

11 passed (13.8s)

## 6. Security / Authorization Tests

Covered in full in `server/tests/lab-03/authorization.api.test.ts` (§2 above, SEC-01 – SEC-09 ).

This category is listed separately here per the course requirement, but the rows are not duplicated — see §2 for the table.

## 7. Migration / Regression Tests

File: `server/tests/lab-03/migration.api.test.ts`

ID	AC	What It Tests	Expected Result	Final
MIG-01	AC-31	Full Lab 2 regression suite (ticket creation, categories, related systems, attachments) re-run using an authenticated Requester session instead of X-Dev-Requester-Id	Every previously-passing Lab 2 case still passes unmodified in behavior	Pass
MIG-02	§7.1 spec	Ticket row count before and after the User migration	Counts match exactly; no ticket is dropped or duplicated	Pass
MIG-03	§7.1 spec	Every migrated Ticket.requesterId resolves to an active User with role REQUESTER corresponding to the original Development Requester identity	100% of migrated tickets resolve correctly	Pass
MIG-04	§7.1 spec	itPriority on every pre-existing ticket after migration	Equals that ticket's requestedPriority (initial backfill rule)	Pass
MIG-05	§5.3 handout	Seed script run twice in sequence against a fresh database	Second run produces identical row counts to the first (idempotent)	Pass

RUN v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

```

✓ tests/lab-03/migration.api.test.ts > MIG-02: Ticket row count before and after the User migr
✓ tests/lab-03/migration.api.test.ts > MIG-03: Every migrated Ticket.requesterId resolves to a
✓ tests/lab-03/migration.api.test.ts > MIG-04: itPriority matches requestedPriority
✓ tests/lab-03/migration.api.test.ts > MIG-05: seed is idempotent
✓ tests/lab-03/migration.api.test.ts > enforces unique and case-insensitive email
✓ tests/lab-03/migration.api.test.ts > MIG-01: preserves authenticated Lab 2 requester workflo
Test Files 1 passed (1)
 Tests 6 passed (6)
Start at 20:49:31
Duration 1.11s (tests 64%, import 21%, transform 15%)

```

## 8. End-to-End Tests

File: e2e/lab-03/authentication.spec.ts

ID	AC	What It Tests	Expected Result	Final
E2E-01	AC-01, AC-08	Full flow: log in as an active user with no pending password change → land on role home → log out → attempt to revisit a protected page	Home screen loads after login; protected page redirects to Login after logout	Pass
E2E-02	AC-02	Initial password login and change	Normal app opens only after a valid change	Pass
E2E-03	AC-06	Attempt login on an inactive account with the correct password	Inactive-account message shown; no access granted	Pass

File: e2e/lab-03/staff-ticket-flow.spec.ts

ID	AC	What It Tests	Expected Result	Final
E2E-04	AC-16, AC-19, AC-21, AC-22	Log in as IT Staff → open Queue → claim an unassigned ticket → set IT Priority → post an Internal Note → transition status to Resolved → log in as the ticket's Requester and confirm the note is not visible	Each step's UI state matches the API result; Requester never sees the Internal Note	Pass
E2E-05	AC-15	Search and filter the Queue end-to-end	Visible rows narrow to match the search/filter combination	Pass

File: e2e/lab-03/user-administration.spec.ts

ID	AC	What It Tests	Expected Result	Final
E2E-06	AC-25, AC-29	Administrator creates a user with an initial password; that user logs in and is forced through Change Password	New user reaches their role home screen only after changing the password	Pass

ID	AC	What It Tests	Expected Result	Final
E2E-07	AC-27, AC-28	Administrator attempts to deactivate their own account, then (as the sole Administrator) attempts to deactivate/role-change themselves via a second seeded admin-adjacent scenario	Both attempts are blocked with the specific inline messages from <code>api-spec.md §5</code>	Pass
E2E-08	AC-09, AC-30	Log in as IT Staff, attempt to visit <code>/admin/users</code> directly by URL	Forbidden state shown; no user data loads	Pass

```
npm notice run software-engineer@1.0.0 test:e2e
npm notice run playwright test e2e/lab-03 responsive.spec.ts
```

Running 11 tests using 1 worker

```
✓ 1 [chromium] › e2e\lab-03\authentication.spec.ts:13:5 › E2E-01: active users can log in a
✓ 2 [chromium] › e2e\lab-03\authentication.spec.ts:23:5 › E2E-02: a temporary password requ
✓ 3 [chromium] › e2e\lab-03\authentication.spec.ts:52:5 › E2E-03: inactive accounts show th
✓ 4 [chromium] › e2e\lab-03\responsive.spec.ts:10:5 › RESP-01: queue switches to its mobile
✓ 5 [chromium] › e2e\lab-03\responsive.spec.ts:18:5 › RESP-02: user panel fills a phone vie
✓ 6 [chromium] › e2e\lab-03\responsive.spec.ts:25:5 › RESP-03: login has no horizontal over
✓ 7 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:11:5 › E2E-04: staff can open a queue
✓ 8 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:38:5 › E2E-05: queue search and statu
✓ 9 [chromium] › e2e\lab-03\user-administration.spec.ts:11:5 › E2E-06: administrator create
✓ 10 [chromium] › e2e\lab-03\user-administration.spec.ts:23:5 › E2E-07: the self-deactivatio
✓ 11 [chromium] › e2e\lab-03\user-administration.spec.ts:33:5 › E2E-08: staff direct navigat
```

11 passed (13.8s)